

So Long: Interactive Storytelling, Embodying Collective Historical Memory, and Participatory Archiving in a VR Voyage

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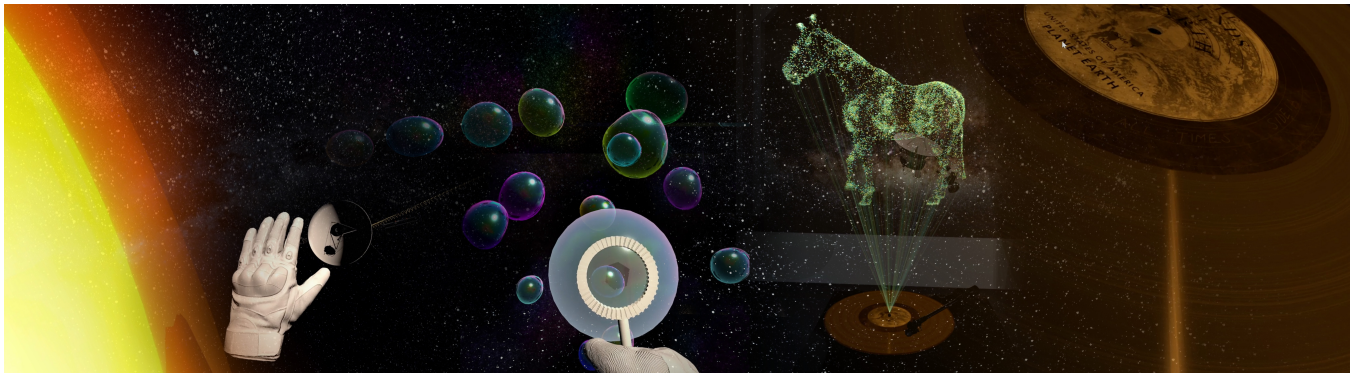


Figure 1: Composite Screenshot from the So Long VR Experience

Abstract

So Long is a VR narrative reimagining Voyager 1's final contact. Users decode the Golden Record, embody the probe, and record messages into an evolving interstellar archive. Through AI-generated environments and embodied interaction, the experience reframes space exploration as participatory memory-making. This paper presents its design, technical approach, and user insights within immersive storytelling and digital heritage.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **Collaborative interaction**; *User studies*; **Scientific visualization**; • **Applied computing** → **Media arts**; **Digital libraries and archives**.

Keywords

immersive storytelling, virtual reality, embodiment, participatory archiving, interactive documentary, digital heritage, user-centred design, co-authorship

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1 Introduction

So Long is a 15-minute interactive VR narrative experience created using Unity and several kinds of generative AI, inspired by NASA's Voyager 1 and the Golden Record. The story centres on the anticipated moment in 2025 when the spacecraft will permanently lose contact with Earth. Drawing from speculative imagination and historical documentation, this piece invites users to relive Voyager's journey and to reflect on humanity's place in the universe [7]. The experience uses embodied interaction and symbolic spatial design to explore memory, empathy, and cosmic connection [9].

Beyond its narrative and artistic scope, So Long also serves as a research project investigating how immersive VR experiences influence users' emotional engagement and comprehension of historical and future narratives [3]. By guiding participants through a poetic journey across time, memory, and civilisation, the project reframes the cosmos not as a distant abstraction but as a shared cultural landscape—one that can be inhabited, perceived, and co-authored through embodied interaction and narrative participation.

2 Artistic Concept and Narrative Structure

The narrative unfolds in four stages: the story begins with a drifting spaceship that stumbles upon the long-lost Voyager 1, where players capture it and begin interacting with its Golden Record. Guided by the onboard AI, they decode visual, auditory, and linguistic content through gamified interactions and holographic projections. Upon decoding a photo of the ocean, players are placed within the image itself, on a metaphorical golden beach, where they discover and cast a message bottle—symbolising humanity's farewell—back into deep space. The final stage visualises the 55 multilingual greetings as a starry sky, in which players not only listen but are also invited to record their own messages. These are embedded into the Golden

Record and launched with Voyager once more, completing a full-circle narrative that transitions players from passive observers to contributors in a collective cosmic archive [4].

A core breakthrough of the project lies in the embodiment within two distinct mini-universes: in the solar system scene, players embody Voyager 1 to traverse its trajectory and experience the probe's first mission—to explore the solar system—from a first-person perspective. In the golden beach scene, players cast a message bottle into space on behalf of humanity, experiencing a collective agency and understanding Voyager's second mission—to carry humanity's legacy into deep space. These two symbolic identities—Voyager and the human collective—are expressed through specific movements and metaphoric environments, reinforcing both cognitive and emotional immersion. Scene transitions further support this duality through audiovisual morphs that link scientific memory with poetic imagination [9].

3 Interaction Design and User Testing

The experience is a 6DoF seated VR project designed for Meta Quest 3, requiring only minimal physical space. The player shifts between observing the Voyager's perspective and interacting as a human contributor to the Golden Record. Gestures include picking up, touching, blowing, and voice input [8].

To ensure usability, over 50 participants from diverse backgrounds took part in iterative user testing. The methodology included semi-structured interviews, in-VR behaviour observation, and post-experience surveys. The experience was described as moving, meditative, and emotionally resonant. Voiceovers were recorded in multiple languages to ensure accessibility and global relatability [5].

4 Technical Framework and Creative Materials

The experience was developed in Unity with XR Interaction Toolkit, featuring Timeline-based scene control and gesture-driven interactivity. Innovations include:

- Depth restoration of NASA imagery using Depth Anything V2 [11], enhancing original archival visuals with inferred 3D structure. This technique provides a layered visual field in VR, allowing users to experience historical imagery with realistic parallax and spatial presence [1].
- Generative visuals and models using Sora [2], Tripo AI [10], and DALL-E [6], used to visualise cosmic environments and conceptual interpretations of Voyager's journey. These AI tools complement historical references with imaginative, humanistic interpretations of interstellar space.
- Custom-designed decoder for Golden Record imagery, enabling participants to gradually reveal encoded images through interaction. This feature translates historical signal data into interactive sequences, encouraging users to engage with archival material as if deciphering an ancient message.
- Procedural spatial audio aligned with planetary transitions is designed to create an embodied sensation of gravitational slingshot effects. Dynamic sound cues synchronised with visual motion blur and depth shifts during planetary flybys simulate the acceleration and inertia of gravitational assists. This induces a visceral 'push-back' effect, enhancing the player's sense of movement as they embody Voyager 1 in

first-person, experiencing the spacecraft's journey through the solar system.

- Holographic visual overlays and dynamic scene orchestration, combining responsive UI elements with real-time environmental transitions. These layers help articulate the temporal and spatial progression of the narrative, reinforcing the metaphorical and factual layers of the Voyager mission.

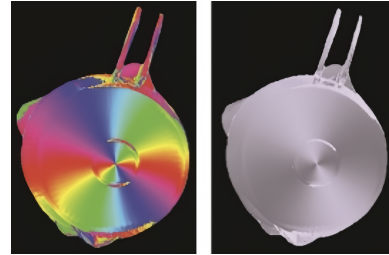


Figure 2: Procedural Generation of Anisotropic Texture Maps

5 Research Contribution

The project investigates how interactive VR can improve narrative transportation, affective engagement, and cognitive retention compared to passive media. Using a mixed-method evaluation framework, including semistructured interviews and behavioral observation, it analyzes how agency, embodied interaction, and participatory authorship contribute to presence and narrative comprehension. A key finding was that the act of leaving behind a personal audio message not only heightened emotional immersion but also fostered a sense of narrative co-ownership and archival continuity. Participants frequently described the experience as "leaving a mark in the universe," demonstrating the potential of VR to elicit reflective and emotionally resonant responses through interactive memory inscriptions.

These results contribute to the ongoing discourse on immersive storytelling, particularly within the domains of experiential learning, participatory memory systems, and virtual heritage.

6 Exhibition and Future Development

So Long has been presented at the AsiaWorld Expo, the China Science Fiction Conference 2025, and academic venues such as University College London, University of Oslo, University of Warsaw and Soochow University. It has also been shortlisted for the Global Awards for World Heritage Education Innovative Cases (AWHEIC), and invited to the London Design Festival and Asia Design Week.

Future directions include multi-user storytelling, biometric-triggered content, and broader cultural expansions of the Golden Record. The team is also developing a shared database to preserve user-recorded messages, transforming the final interaction into a form of participatory archiving. This system allows the voice of each player to become part of a growing collective narrative, positioning the experience as an interactive documentary and a co-created cultural artifact. WebXR deployment is also being explored to improve accessibility.

By evoking poetic imagery and encouraging user reflection through interaction, So Long reimagines how human memory might persist in space and time.

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